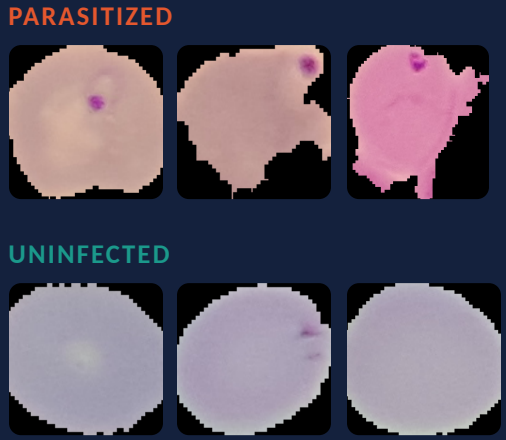
CSA431 COMPUTER VISION · M4 + M5

Does cell-weighted sensitivity hide hard Patient-IDs?



M4: one pre-registered evaluation change, plus ablations

M5: failure analysis, IEEE paper and defence

Dataset: NIH / NLM thin-smear cells, 27,558 images · Base paper: Rajaraman et al., 2018, PeerJ 6:e4568

Presenters: Udita (230082) · Aman Kumar (230064)

Rishihood University

MOTIVATION

The question M4 answers

98.6%
accuracy, cell-level split
(Rajaraman, Table 6)

→

95.9%
accuracy, patient-level split
(Rajaraman, Table 6)

- Same-source cells share stain, lighting and focus, so cells are not independent.
- A pooled, cell-weighted mean can be carried by a few large Patient-IDs.
- Course rules: predict first; report sensitivity + AUC; no clinical claims.

OPEN QUESTION

Under leakage-free folds, does a pooled mean hide Patient-IDs with concentrated false negatives?

Answered by re-reading the frozen M2 predictions. No new model, no retraining.

M2 RECAP

What we reproduced

94.24%	0.9832	93.90%	93.56%
sensitivity	AUC	accuracy	specificity
paper 94.7% · Δ −0.46 pp	paper 0.991 · Δ −0.0078	paper 95.9% · Δ −2.00 pp	paper 97.2% · Δ −3.64 pp

Frozen ResNet-50
ImageNet weights

→

2,048 features
224×224 input

→

FC 2048→512→1
dropout 0.5, 20 epochs

→

Threshold 0.5
Parasitized = positive

Five folds grouped by the official NLM Patient-ID key (201 IDs), zero train/test overlap, seed 42.

Approximate reproduction: closest on sensitivity, lower on specificity. The paper's folds, feature layer and search were not available to us. These frozen out-of-fold predictions are the only input to M4 and M5.

Hypothesis and decision rule

Saved before any M4 code · first committed in **f3152be**, 03 Oct 2026 23:48 +0530

IF	the frozen M2 predictions are summarised per official Patient-ID within each fold
THEN	macro sensitivity will be lower than pooled sensitivity, by about 1 to 3 pp
BECAUSE	large IDs dominate a cell-weighted mean, so hard small IDs can disappear inside it

Supported
≥ 4 of 5 folds negative **and** mean ≤ −1.68 pp

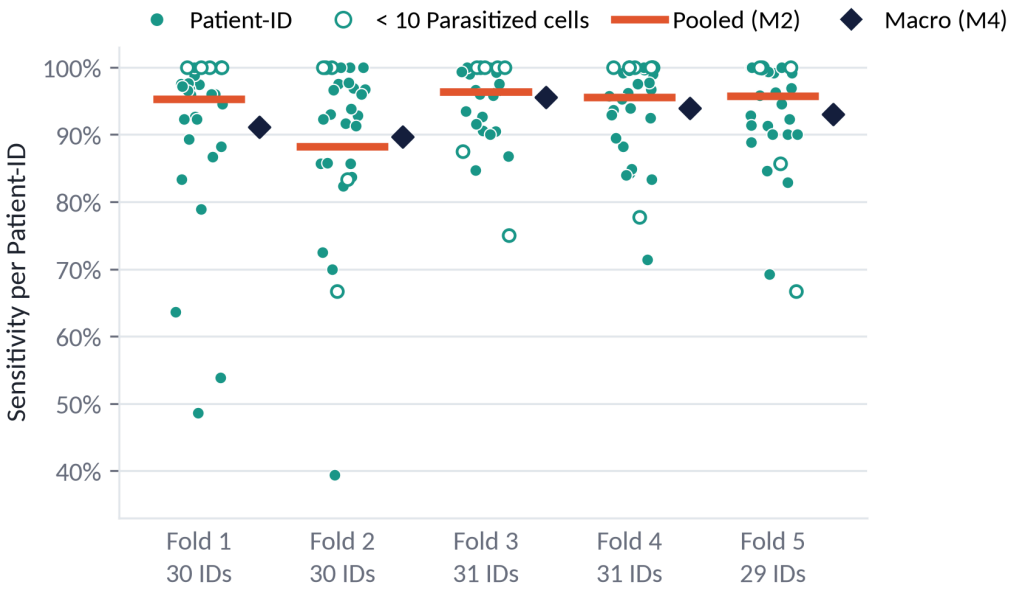
Unsupported
≥ 4 of 5 folds positive **and** mean ≥ +1.68 pp

Inconclusive
anything else, including a smaller negative mean

−1.68 pp is half the M2 sensitivity fold SD (3.36 pp). Macro uses the 151 IDs with Parasitized cells; the other 50 are counted, not dropped.

METHOD · M4

One change: how the same rows are averaged



POOLED · M2

Every **cell** counts once:
ΣTP / Σ(TP + FN)

MACRO · M4

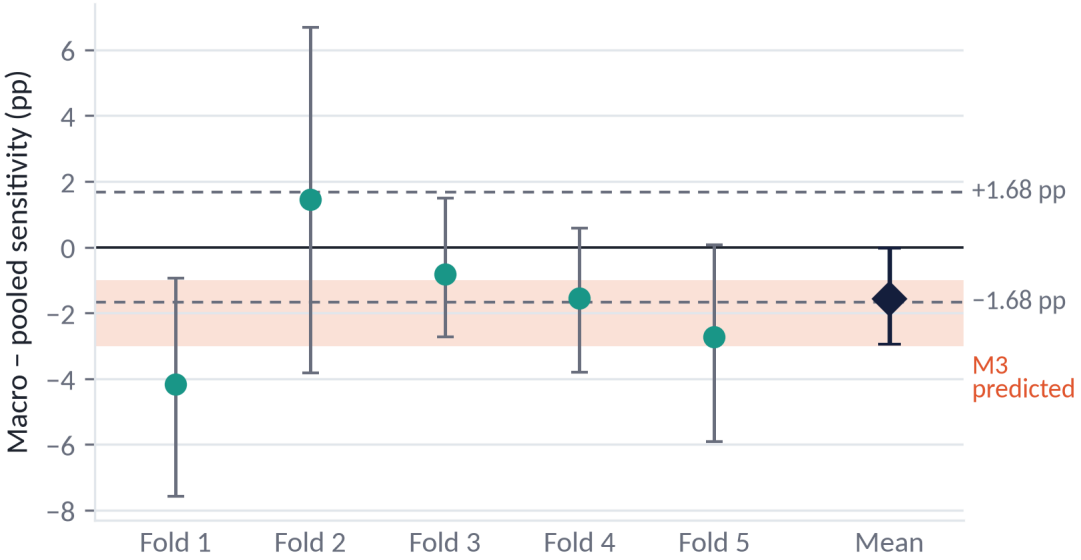
Every **Patient-ID** counts once:
mean of $TP_i / (TP_i + FN_i)$

- ✓ Model, features, probabilities fixed
- ✓ 0.5 threshold, labels, folds fixed
- ≠ **Only the aggregation changes**

Reuse checks PASS: 27,558 rows, folds identical, pooled metrics match to 1.1e-16

PRIMARY RESULT

Right direction, short of the threshold



PREDICTED (M3)

−1 to −3 pp

mean macro − pooled

OBSERVED (M4)

−1.56 pp

4 of 5 folds negative · inside the range

VERDICT · M3 RULE

INCONCLUSIVE

mean must be ≤ −1.68 pp; short by 0.12 pp

Mean: 95% cluster-bootstrap CI [−2.95, −0.04] pp · across-fold t-interval [−4.18, 1.05] pp · n = 5 folds, so no significance claims

ABLATIONS · EXPLORATORY

The verdict is fragile, not reversed

A1 · WEIGHTING

= pooled

Weighting each ID by its cell count gives back the pooled value exactly: the gap *is* the weighting.

A2 · ELIGIBILITY

−2.00 / −1.28 pp

Mean with ≥ 10 / ≥ 25 Parasitized cells per ID: one variant passes the threshold, one does not.

A3 · LEAVE ONE ID OUT

2 of 151

single-ID deletions flip the verdict: C116P77ThinF (fold 2), C182P143NThinF (fold 5).

A4 · CLUSTER BOOTSTRAP

[−2.95, −0.04] pp

95% CI for the mean (2,000 resamples): excludes 0, includes −1.68 pp.

A5 · SPECIFICITY

+0.01 pp

Every ID has 63–77 Uninfected cells, so weighting cannot move specificity.

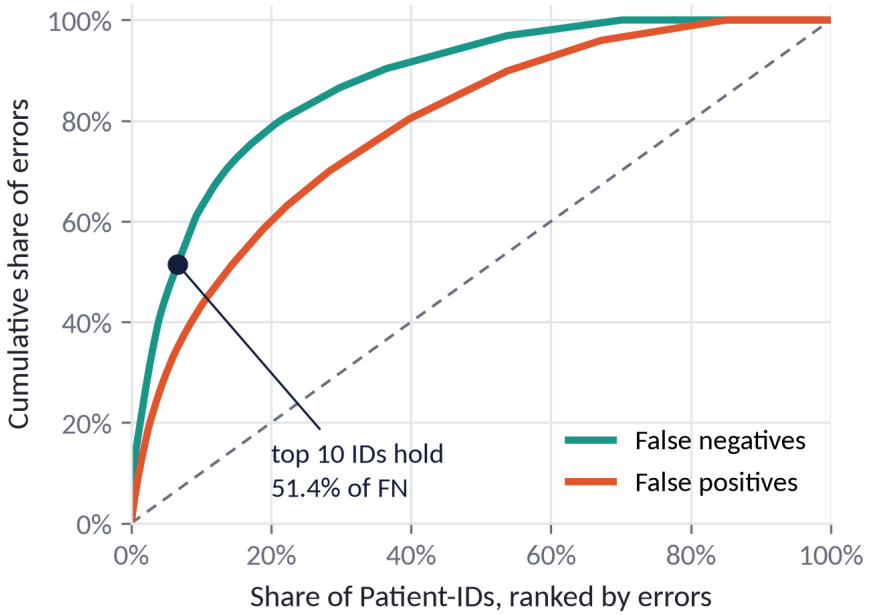
TAKEAWAY

Planned before computing; exploratory only.

Pre-registered verdict stands

INCONCLUSIVE

False negatives cluster in a few Patient-IDs



51.4%

of 795 false negatives sit in the top 10 Patient-IDs; 10 IDs hold half. False positives are more diffuse (top 10: 29.7%).

Fold 2

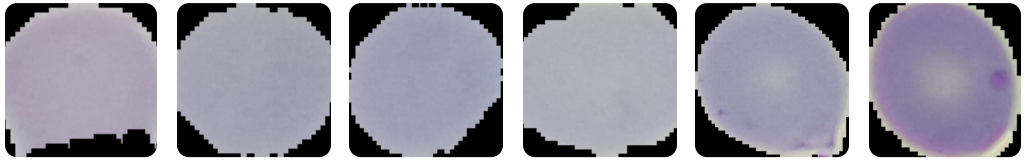
325 FN and 6 of the 10 worst IDs; C116P77ThinF alone holds 35.1%. Large, **hard** IDs pull pooled *below* macro.

28.7%

of FN are confident ($p < 0.1$); 24.5% of FP have $p > 0.9$. Brier 0.0469, ECE 0.0116.

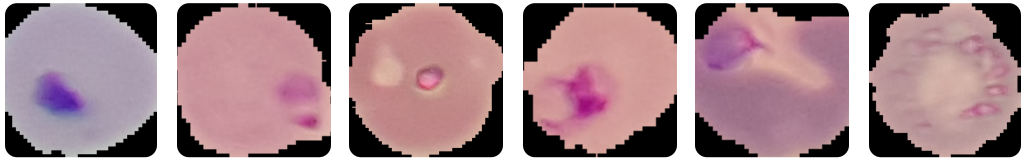
The most confident errors may be label questions

Most confident false negatives · labelled Parasitized



p = 0.004 p = 0.004 p = 0.004 p = 0.005 p = 0.005 p = 0.006

Most confident false positives · labelled Uninfected



p = 1.000 p = 1.000 p = 1.000 p = 1.000 p = 1.000 p = 1.000

Official, hash-verified crops; resized, colours unaltered. p = model probability of Parasitized.

Crops with a visible stained inclusion



Blind review of 96 crops by **one non-expert reviewer** (the AI assistant), categories fixed before viewing. **Observation, not cause:** label noise, faint parasites, border debris or reviewer error.

LIMITATIONS

What this result cannot tell you

1 n = 5 folds

One grouping seed, so no significance claims. Fold signs could change with another grouping.

2 Fragile verdict

Near the threshold; one ID or one eligibility rule moves it. Variants are not adopted.

3 Patient-ID ≠ slide

An official mapping key; one source patient has two keys. Not a patient diagnosis.

4 Approximate M2

Specificity −3.64 pp vs published; failure modes describe this model, not paper1's.

5 Image review

One non-expert reviewer, small crops, errors selected by model confidence.

6 Scope

One public segmented-cell dataset, no external data. **No clinical claim.**

NEXT STEPS

What we would do next

1

Repeat the split

Several grouping seeds: how often does a fold flip sign?

2

Expert re-review

A microscopist labels the confident errors, with inter-rater agreement.

3

Stain-aware check

Do grey/blue-tinged crops come from particular slides?

4

Recalibration

Only as its own change, with its own prediction.

EACH STEP IS A NEW INTERVENTION

Whatever comes next: report pooled *and* per-Patient-ID sensitivity side by side, with the group distribution.

CONCLUSION

What M4 and M5 show

1. Rajaraman's patient-level ResNet-50 baseline reproduced approximately: sensitivity 94.24%, AUC 0.9832.
2. One pre-registered evaluation change: right direction in 4/5 folds and inside the predicted range, but short of the threshold.
3. Whether cell weighting hides or exaggerates hard groups depends on **which groups are large**. The most confident errors may be label questions.

PRE-REGISTERED VERDICT

INCONCLUSIVE

mean −1.56 pp
rule: ≤ −1.68 pp

Thank you · questions welcome

Public NIH/NLM segmented-cell data; no clinical claims · paper: paper/main.pdf